

Code-Layout, Readability and Reusability

Date	01 July 2026
Team ID	SWTID-2026-3278
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation maintained throughout the Python code.
2	Proper File Structure	Files and folders are logically organized	Yes	Files organized into dataset, model, templates, and static folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Clear variable names such as income, credit_score, age.
4	Function / Method Names	Functions are descriptively named	Yes	Functions named clearly, e.g., predict_approval() and train_model().
5	Code Comments	Inline and block comments explain logic	Yes	Comments added to explain important sections of the code.
6	Modular Design	Code is split into reusable functions/modules	Yes	Code divided into reusable modules for preprocessing and prediction.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Unused and duplicate code removed.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling implemented.

Reusable Components / Modules:

S. No	Component / Module Name	Language /Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing	Python	Model Training	Cleans and prepares data consistently.

2	ML Prediction Model	Scikit-learn	Prediction Module	Reusable for all new applications.
3	Flask Application	Flask	Web Interface	Provides user interaction and prediction.
4	Dataset Loader	Pandas	Training & Testing	Efficient data loading and processing.
5	User Interface	HTML, CSS	Frontend	Easy maintenance and future updates.

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure.
Readability	5	Easy-to-read code with meaningful names.
Reusability	5	Modular and reusable components.
Documentation / Comments	4	Sufficient comments for understanding.
Overall Score	4.8 / 5	Clean, maintainable, and reusable code.